

## 1. General Description

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The EMS3580 is a Dual SPDT analog switch with ultra-low distortion, high OFF-Isolation for special stereo audio applications with negative swing audio signals capacity that features ultra-low  $R_{on}$  of  $0.16\Omega$  (typical) at  $3.3V V_{CC}$ .

The EMS3580 operates a single power supply over a wide range from  $3.0V$  to  $4.5V$  and  $1.8V$  logic compatible with high PSRR. With a special pop-click shunt circuitry at each signal pin that eliminates pops and clicks associated at any application conditions likes switched, enable/disable, power-down and power-up.

With superior THD+N performance and other high performance, the EMS3580 is an ideal device for Hi-Fi system applications.

The EMS3580 is available in QFN1826-16L package. All product is Pb-free and Halogen-free.

## 2. Features and Benefits

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- Single supply range operating from  $3.0V$  to  $4.5V$
- $-118dB$  THD+N into  $100k\Omega$  load at  $2V_{rms}$
- $-114dB$  THD+N into  $32\Omega$  load at  $2V_{rms}$
- Signal-to-Noise (SNR) Ratio  $132dBA$
- $100dB$  PSRR at  $10kHz$
- $145dB$  crosstalk & separation
- Pop/Click shunt circuit
- Audio Path Soft Turn-On/Off for Pop & Click Elimination

## 3. Applications

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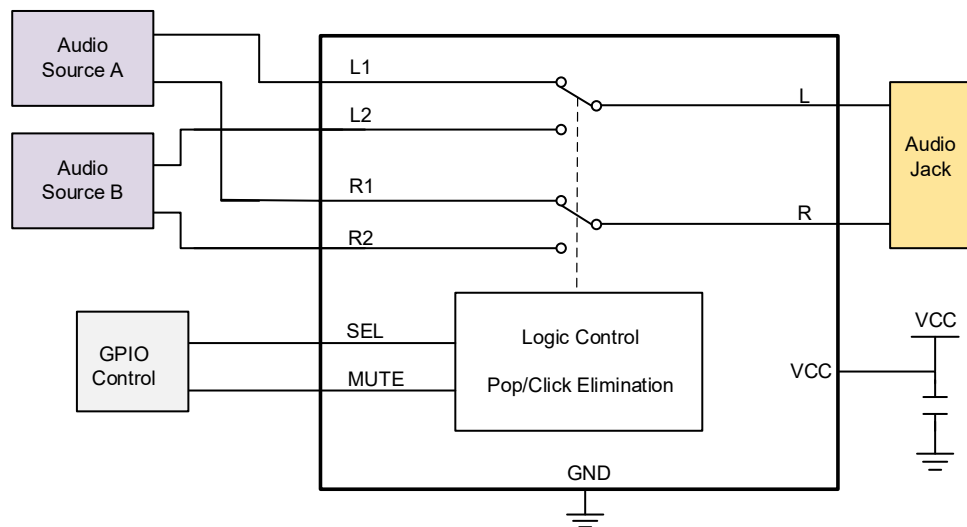
- Hi-Fi Smartphones and Portable Device
- Hi-Fi SACD/DVD Players
- High Quality Home Theaters

## 4. Ordering Information

**Table 1. Ordering information**

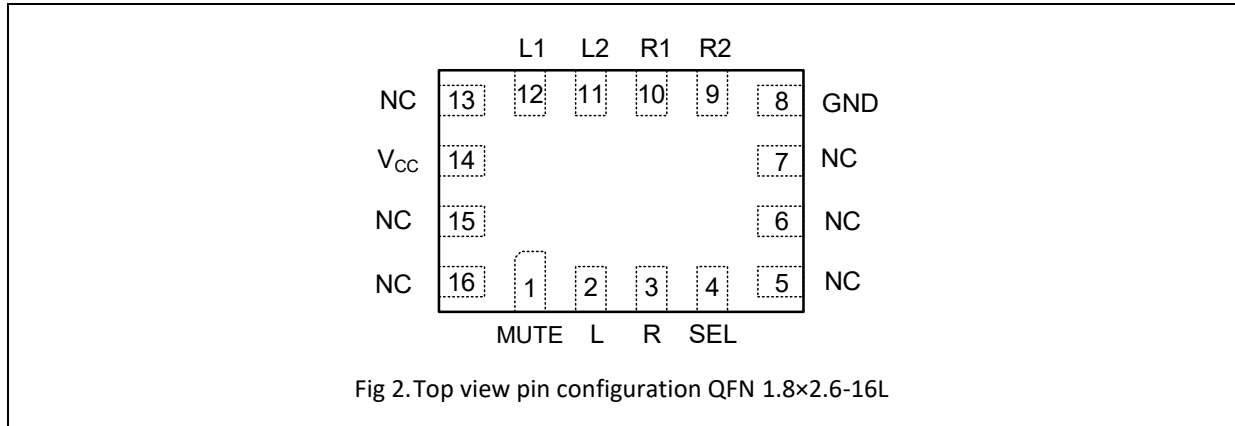
| Type number | Package        |                                                              |          |
|-------------|----------------|--------------------------------------------------------------|----------|
|             | Name           | Description                                                  | Quantity |
| EMS3580RSV  | QFN1.8×2.6-16L | QFN package, 16 pins<br>1.8 mm × 2.6 mm; 0.6 mm (Max) height | 3000     |

## 5. Functional Diagram


**Fig 1. Functional diagram**

## 6. Pinning Information

### 6.1. Pin map



### 6.2. Pin description

Table 2. Pin description

| Symbol          | Pin                 | Description                                    |
|-----------------|---------------------|------------------------------------------------|
| L, R            | 2, 3                | Left and right channel Common port             |
| R1, L1          | 10, 12              | Left and right channel port 1, normally closed |
| R2, L2          | 9, 11               | Left and right channel port 2, normally open   |
| MUTE            | 1                   | Enable control, active high                    |
| SEL             | 4                   | Port selection control pin                     |
| V <sub>CC</sub> | 14                  | Power supply                                   |
| GND             | 8                   | Ground                                         |
| NC              | 5, 6, 7, 13, 15, 16 | Not connection                                 |

## 7. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; X = Don't care.

| Input |     | Switch |        |
|-------|-----|--------|--------|
| MUTE  | SEL | L1, R1 | L2, R2 |
| L     | L   | ON     | OFF    |
| L     | H   | OFF    | ON     |
| H     | X   | OFF    | OFF    |

## 8. Absolute Maximum Ratings

Absolute Maximum Ratings may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions beyond those indicated in the operational sections of this specification is not implied.

**Table 4. Absolute Maximum Ratings**

| Symbol                              | Parameter                                                                          | Min  | Max   | Unit |
|-------------------------------------|------------------------------------------------------------------------------------|------|-------|------|
| V <sub>CC</sub>                     | supply voltage                                                                     | -0.3 | 5.5   | V    |
| V <sub>IN</sub>                     | digital control input voltage                                                      | -0.3 | 5.5   | V    |
| V <sub>IS</sub>                     | analog input/output voltage (L1, L2, R1, R2, L, R)                                 | -4.0 | 4.0   | V    |
| I <sub>IO</sub>                     | switch continuous current (L1, L2, R1, R2, L, R)                                   |      | ±300  | mA   |
| I <sub>IO_PK</sub>                  | switch peak current (L1, L2, R1, R2, L, R)<br>(pulsed at 1ms, 10% duty cycle, Max) |      | ±500  | mA   |
| P <sub>D</sub>                      | power dissipation in still air                                                     |      | 250   | mW   |
| T <sub>STG</sub>                    | storage temperature range                                                          | -55  | 150   | °C   |
| T <sub>J</sub>                      | junction temperature                                                               |      | 150   | °C   |
| T <sub>L</sub>                      | lead temperature (Soldering, 10 seconds)                                           |      | 260   | °C   |
| R <sub>θJA</sub>                    | thermal resistance                                                                 |      | 80    | °C/W |
| I/O to V <sub>CC</sub> , I/O to GND | ESD protection (HBM)                                                               |      | ±6000 | V    |
| I/O to I/O                          |                                                                                    |      | ±4000 | V    |

## 9. Recommended Operating Conditions

**Table 5. Recommend operating ratings**

| Symbol          | Parameter                                          | Min  | Max             | Unit |
|-----------------|----------------------------------------------------|------|-----------------|------|
| V <sub>CC</sub> | supply voltage                                     | 3.0  | 4.5             | V    |
| V <sub>IN</sub> | digital control input voltage                      | 0    | V <sub>CC</sub> | V    |
| V <sub>IS</sub> | analog input/output voltage (L1, L2, R1, R2, L, R) | -3.3 | V <sub>CC</sub> | V    |
| T <sub>A</sub>  | operating temperature                              | -40  | 85              | °C   |

## 10. Electrical Characteristics

**Table 6. Electrical characteristics**
 $T_a = 25\text{ }^{\circ}\text{C}$ ,  $V_{CC}=3.6\text{V}$ ,  $V_{IS}=2V_{rms}$ ,  $R_L=32\Omega$ ,  $f=1\text{kHz}$ , unless otherwise noted

| Symbol                               | Parameter                                     | Conditions                                                                                                                              | Min | Typ         | Max | Unit      |
|--------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----|-------------|-----|-----------|
| <b>Analog Switch Characteristics</b> |                                               |                                                                                                                                         |     |             |     |           |
| $V_{IS}$                             | analog signal range                           | $V_{DD} = 3.3\text{ V to }4.2\text{ V}$                                                                                                 |     | 2.5         |     | $V_{rms}$ |
| $R_{ON}$                             | on-resistance                                 | $V_{IS} = -3.3\text{ V to }3.3\text{V}$<br>$I_{OUT} = 100\text{ mA}$                                                                    |     | 0.16        |     | $\Omega$  |
| $\Delta R_{ON}$                      | $R_{ON}$ matching between channels [1]        | $V_{IS} = -3.3\text{ V to }3.3\text{V}$<br>$I_{OUT} = 100\text{ mA}$                                                                    |     | 0.007       |     | $\Omega$  |
| $R_{FLAT(ON)}$                       | $R_{ON}$ flatness [2]                         | $V_{IS} = -3.3\text{ V to }3.3\text{V}$<br>$I_{OUT} = 100\text{ mA}$                                                                    |     | 0.001       |     | $\Omega$  |
| <b>Dynamic Characteristics</b>       |                                               |                                                                                                                                         |     |             |     |           |
| THD+N                                | total harmonic distortion                     | $f = 10\text{ Hz to }22\text{ kHz}$<br>$V_{IS} = 2\text{ }V_{rms} @ R_L = 100\text{ k}\Omega$                                           |     | -118        |     | dB        |
|                                      |                                               | $f = 10\text{ Hz to }22\text{ kHz}$<br>$V_{IS} = 2\text{ }V_{rms} @ R_L = 32\text{ }\Omega$                                             |     | -114        |     | dB        |
|                                      |                                               | $f = 10\text{ Hz to }500\text{ kHz}$<br>$V_{IS} = 1.55\text{ }V_{rms} @ R_L = 100\text{ k}\Omega$                                       |     | -104        |     | dB        |
| IMD                                  | intermodulation distortion                    | Mode = CCIF 19 k + 20 k<br>Ratio = 1<br>$V_{IS} = 500\text{ m}V_{rms} @ R_L = 100\text{ k}\Omega$                                       |     | -122        |     | dB        |
|                                      | dynamic/transient intermodulation distortion  | Mode = DIM 100<br>$V_{IS} = 1\text{ }V_{rms} @ R_L = 100\text{ k}\Omega$                                                                |     | -103        |     | dB        |
| SNR                                  | signal-to-noise ratio                         | $f = 10\text{ Hz to }22\text{ kHz}$ ,<br>inputs grounded<br>$R_L = 32\text{ }\Omega$ or $100\text{ k}\Omega$                            |     | 132         |     | dBA       |
| IMB                                  | stereo channel imbalance L1 and R1, L2 and R2 | $f = 10\text{ Hz to }22\text{ kHz}$ ,<br>$R_L = 100\text{ k}\Omega$                                                                     |     | $\pm 0.003$ |     | dB        |
| OIRR                                 | off isolation (Muting)                        | $f = 10\text{ Hz to }22\text{ kHz}$ ,<br>$V_L = V_R = 2\text{ }V_{rms} @ R_L = 100\text{ k}\Omega$<br>MUTE = $V_{CC}$ , SEL = "X"       |     | 145         |     | dB        |
|                                      |                                               | $f = 10\text{ Hz to }22\text{ kHz}$ ,<br>$V_{LX} = V_{RX} = 2\text{ }V_{rms} @ R_L = 100\text{ k}\Omega$<br>MUTE = $V_{CC}$ , SEL = "X" |     | 145         |     | dB        |

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## High Fidelity Stereo SPDT Switch with Pop and Click Elimination

| Symbol                        | Parameter                          | Conditions                                                                                                                                   | Min | Typ | Max | Unit |
|-------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------|
|                               |                                    | f = 10 Hz to 22 kHz,<br>V <sub>L</sub> = V <sub>R</sub> = 2 V <sub>rms</sub> @ R <sub>L</sub> = 32 Ω<br>MUTE = V <sub>CC</sub> , SEL = "X"   |     | 127 |     | dB   |
|                               |                                    | f = 10 Hz to 22 kHz,<br>V <sub>Lx</sub> = V <sub>Rx</sub> = 2 V <sub>rms</sub> @ R <sub>L</sub> = 32 Ω<br>MUTE = V <sub>CC</sub> , SEL = "X" |     | 127 |     | dB   |
| X <sub>talk</sub>             | crosstalk (Channel-to-Channel) [3] | f = 10 Hz to 22 kHz,<br>V <sub>IS</sub> = 2 V <sub>rms</sub> ,<br>source impedance = 0 Ω<br>R <sub>L</sub> = 100 kΩ                          |     | 145 |     | dB   |
| PSRR                          | power supply ripple rejection      | f = 10 kHz,<br>V <sub>IS</sub> = 0.1 V <sub>rms</sub> ,<br>inputs grounded                                                                   |     | 100 |     | dB   |
| BW                            | -3dB bandwidth                     | R <sub>L</sub> = 50 Ω                                                                                                                        |     | 50  |     | MHz  |
| T <sub>TRS-OM</sub>           | on-to-mute time                    |                                                                                                                                              |     | 57  |     | ms   |
| T <sub>TRS-MO</sub>           | mute-to-on time                    |                                                                                                                                              |     | 35  |     | ms   |
| T <sub>OFF</sub>              | turn-off time                      | V <sub>IS</sub> =1.5V, R <sub>L</sub> = 20kΩ<br>MUTE=0                                                                                       |     | 3.5 |     | us   |
| T <sub>ON</sub>               | turn-on time                       | V <sub>IS</sub> =1.5V, R <sub>L</sub> = 20kΩ<br>MUTE=0                                                                                       |     | 47  |     | ms   |
| T <sub>BBM</sub>              | break-before-make time             | V <sub>IS</sub> =1.5V, R <sub>L</sub> = 20kΩ<br>MUTE=0                                                                                       |     | 45  |     | ms   |
| C <sub>OFF</sub>              | Lx, Rx off capacitance             | f=100kHz,<br>V <sub>Lx</sub> or V <sub>Rx</sub> = V <sub>L</sub> or V <sub>R</sub> = 0 V                                                     |     | 56  |     | pF   |
| C <sub>ON</sub>               | L, R on capacitance                | f=100kHz,<br>V <sub>Lx</sub> or V <sub>Rx</sub> = V <sub>L</sub> or V <sub>R</sub> = 0 V                                                     |     | 84  |     | pF   |
| Power Supply Characteristics  |                                    |                                                                                                                                              |     |     |     |      |
| I <sub>CC</sub>               | supply quiescent current           | MUTE = 0 V                                                                                                                                   |     | 160 |     | μA   |
|                               |                                    | MUTE = V <sub>CC</sub>                                                                                                                       |     | 80  |     | μA   |
| Digital Input Characteristics |                                    |                                                                                                                                              |     |     |     |      |
| V <sub>IH</sub>               | digital input logic high level     | V <sub>CC</sub> = 3.6 V to 4.5 V                                                                                                             | 1.6 |     |     | V    |
|                               |                                    | V <sub>CC</sub> = 3 V to 3.6 V                                                                                                               | 1.5 |     |     | V    |
| V <sub>IL</sub>               | digital input logic low level      | V <sub>CC</sub> = 3.6 V to 4.5 V                                                                                                             |     |     | 0.5 | V    |

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### High Fidelity Stereo SPDT Switch with Pop and Click Elimination

| Symbol   | Parameter                     | Conditions                             | Min | Typ | Max       | Unit             |
|----------|-------------------------------|----------------------------------------|-----|-----|-----------|------------------|
|          |                               | $V_{CC} = 3\text{ V to } 3.6\text{ V}$ |     |     | 0.4       | V                |
| $I_{IN}$ | digital input leakage current |                                        |     |     | $\pm 2.0$ | $\mu\text{A}$    |
| $R_{PD}$ | SEL pull-down resistor        |                                        |     | 4   |           | $\text{M}\Omega$ |
| $R_{PU}$ | MUTE pull-up resistor         |                                        |     | 43  |           | $\text{M}\Omega$ |

[1]  $R_{ON}$  matching between channels is calculated by subtracting the channel with the lowest max  $R_{ON}$  value from the channel with the highest max  $R_{ON}$  value.

[2] Flatness is defined as the difference between maximum and minimum value of ON-resistance at the specified analog signal voltage points.

[3] Crosstalk is inversely proportional to source impedance.

## 11. Typical Performance Curves

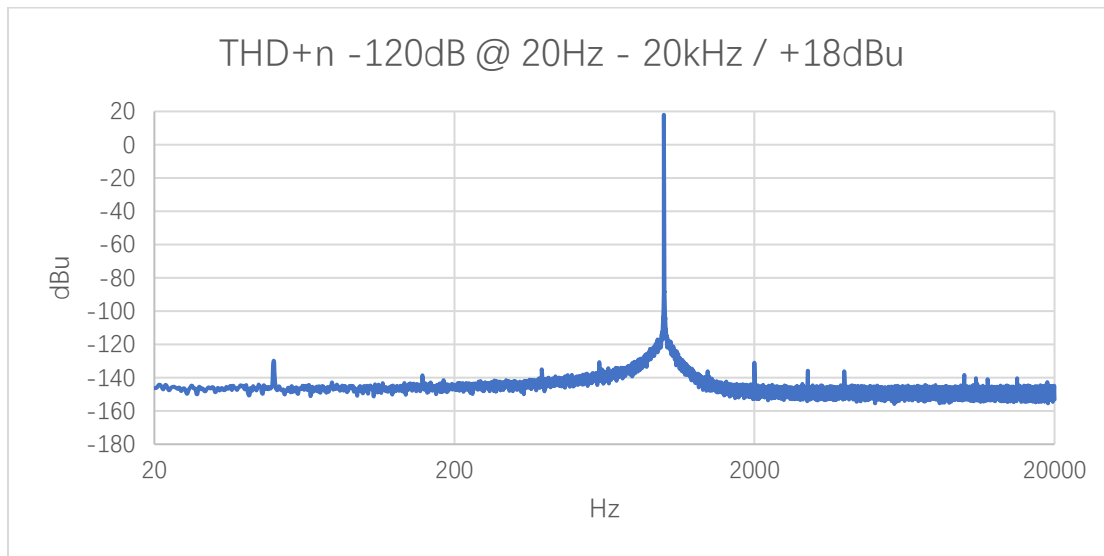


Fig 3. Typical THD+n / 1kHz Tone @ 20Hz - 20kHz (Measurement Limit)

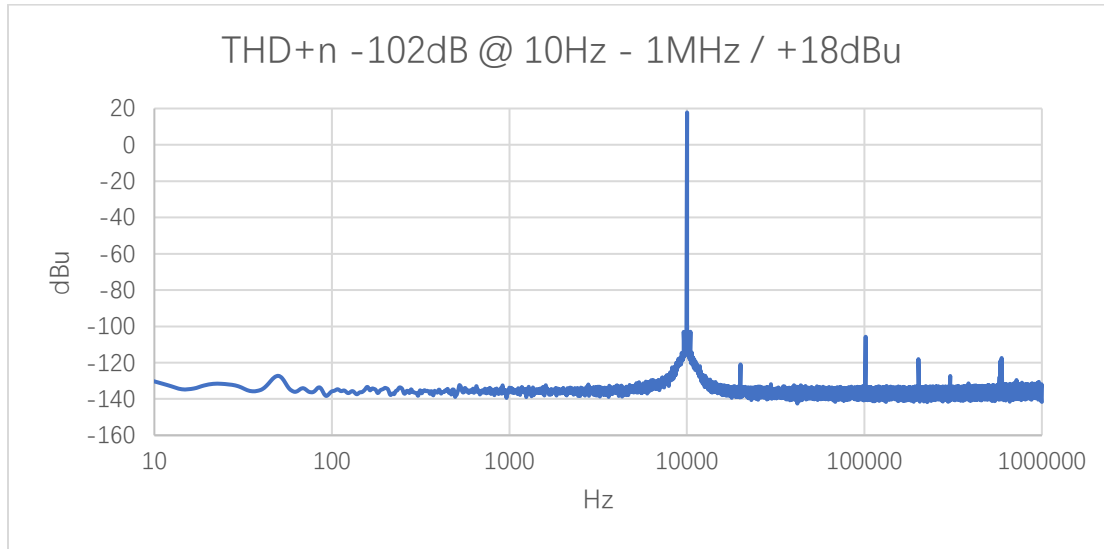


Fig 4. Typical THD+n / 10kHz Tone @ 10Hz - 1MHz (Measurement Limit)

## 11.1 Source Impedance

Higher source impedance will degrade THD performance, so please design the source circuit carefully, add front buffer circuit for lowest source impedance is good solution generally.

## 11.2 Signal Swing

The EMS3580 design for +18 dBu(6.15Vrms) Differential audio system. The damage limit is +20.5dBu(8.2Vrms) Differential, >2dB safe margin is retained.

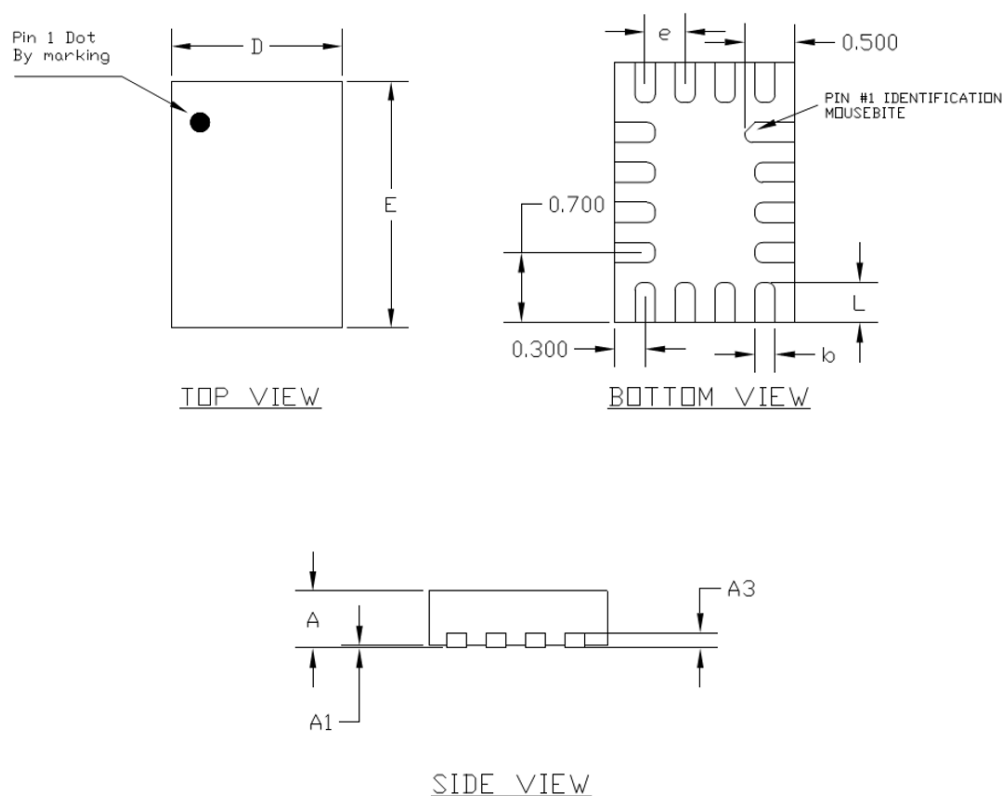
**Table 7. Recommend Signal Swing Amplitude**

| VCC = 3.3V                | VCC = 3.6V                | VCC = 3.9V                | VCC = 4.2V                | VCC = 4.5V                |
|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| 2.3 V <sub>rms</sub> SE   | 2.4 V <sub>rms</sub> SE   | 2.7 V <sub>rms</sub> SE   | 2.9 V <sub>rms</sub> SE   | 3.1 V <sub>rms</sub> SE   |
| 4.6 V <sub>rms</sub> DIFF | 4.8 V <sub>rms</sub> DIFF | 5.4 V <sub>rms</sub> DIFF | 5.8 V <sub>rms</sub> DIFF | 6.2 V <sub>rms</sub> DIFF |



## 12. Package Outline

### QFN 1.8x2.6 -16L



| Symbol | Dimension in Millimeters |      |      |
|--------|--------------------------|------|------|
|        | Min.                     | Typ. | Min. |
| A      | 0.50                     | 0.55 | 0.60 |
| A1     | 0.00                     | -    | 0.05 |
| A3     | 0.15 Typ.                |      |      |
| D      | 1.75                     | 1.80 | 1.85 |
| E      | 2.55                     | 2.60 | 2.65 |
| L      | 0.30                     | 0.40 | 0.50 |
| b      | 0.15                     | 0.20 | 0.25 |
| e      | 0.40 Typ.                |      |      |

## 12. Abbreviations

**Table 8. Abbreviations**

| Acronym | Description             |
|---------|-------------------------|
| ESD     | ElectroStatic Discharge |
| HBM     | Human Body Model        |

## 13. Revision History

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**Table 9. Revision history**

| Document ID      | Release Date | Data sheet status | Change notice | Supersedes |
|------------------|--------------|-------------------|---------------|------------|
| EMS3580 Rev. 1.0 | Feb 21, 2026 | Product datasheet |               |            |